STATISTICS 652

LECTURE 7 - 2026

MULTIVARIATE DISTRIBUTIONS: $\underset{\sim}{X} \in \mathbb{R}^2$ RANDOM VECTOR - REVIEW

PROBABILITY MEASURE: $P_X(B) = P(\underset{\sim}{X} \in B)$ $B \subset \mathbb{R}^2$ (BOREL SUBSET)

DISTRIBUTION FUNCTION: $F_X(\underset{\sim}{x}) = P(X_1 \le x_1, \ldots, X_2 \le x_2)$, $\underset{\sim}{x} \in \mathbb{R}^2$

EQUIVALENCE: $P_X = P_Y \iff F_X = F_Y$

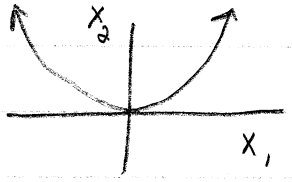
JORDAN DECOMPOSITION:

$F = F_d + F_c$ DISCRETE AND CONTINUOUS COMPONENTS

$= F_d + F_{ac} + F_s$ CONTINUOS: ABSOLUTELY CONTINUOUS AND SINGULAR COMPONENTS: $F_c = F_{ac} + F_s$

EXAMPLES:

1) $\underset{\sim}{X} = \begin{pmatrix} X \\ X^2 \end{pmatrix}$ with $X \sim N(0,1)$



SUPPORT SET

CONTINUOUS SINCE THERE IS NO POINT MASS

NO ABSOLUTELY CONTINUOUS COMPONENT SINCE $\underset{\sim}{X}$ DOES NOT HAVE A DENSITY IN $\mathbb{R}^2$. THAT IS, THERE EXIST A $B \subset \mathbb{R}^2$ such that $P(X \in B) = 1$ and LEBESGUE MEASURE $(B) = 0$.

2) P_n, F_n: empirical measure AND EMPIRICAL DISTRIBUTION of $\{\underset{\sim}{x}_1, \ldots, \underset{\sim}{x}_n\}$ where $\underset{\sim}{x}_i \in \mathbb{R}^2$, $i = 1, \ldots, n$

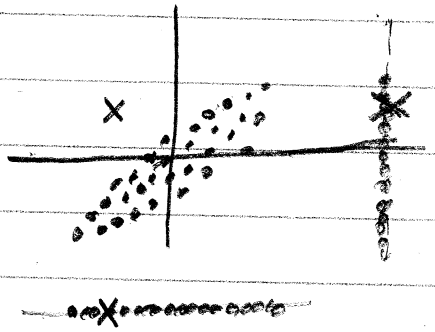
ALWAYS DISCRETE with $P(\underset{\sim}{X} = \underset{\sim}{x}_i) = \frac{1}{n}$ $i = 1, \ldots, n$.

SELF REVIEW: MARGINALS, JOINT MARGINALS, CONDITIONAL DISTRIBUTIONS, INDEPENDENCE.

COMMENT: $\underset{\sim}{X} \in \mathbb{R}^q$ and $\underset{\sim}{Y} \in \mathbb{R}^q$ CAN HAVE ALL THE SAME p-dimensional MARGINAL DISTRIBUTIONS FOR $p = 1, 2, \ldots, q-1$, BUT THIS DOES NOT IMPLY $\underset{\sim}{X}$ and $\underset{\sim}{Y}$ HAS THE SAME DISTRIBUTION.

COMMENT: AN OUTLIER IN a q-dimensional data set MAY NOT BE ANY OUTLIER IN ANY p-dimensional MARGINAL DISTRIBUTION, $p = 1, 2, \ldots, q-1$.

e.g. F_n



CANNOT IDENTIFY THE OUTLIER X FROM ONLY THE 2 MARGINALS.

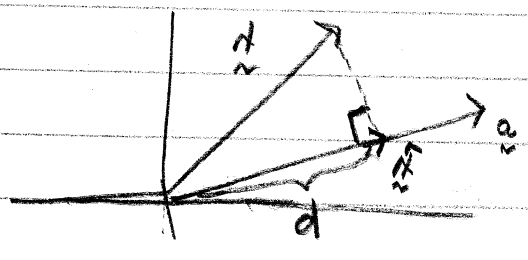
NOTE: IN HIGHER DIMENSIONS, OUTLIERS CAN BE VERY DIFFICULT TO DETECT.

NOTE: OUTLIERS, THOUGH, CAN BE SEEN IN SOME UNIVARIATE PROJECTION OF THE DATA. THIS GIVES RISE TO THE AREA OF <u>PROJECTION PURSUIT</u>.

LINEAR COMBINATION. $\underset{\sim}{a} : \mathbb{R}^q \to \mathbb{R}$ with $y = \sum_{j=1}^{q} a_j x_j = \underset{\sim}{a}' \underset{\sim}{x}$ $\underset{\sim}{a} \in \mathbb{R}^q$

UNIVARIATE ORTHOGONAL PROJECTION:

NOTE: $d = \dfrac{\underset{\sim}{a}' \underset{\sim}{x}}{\| \underset{\sim}{a} \|}$



$$d = \frac{\underset{\sim}{a}'\underset{\sim}{x}}{\|\underset{\sim}{a}\|} = \underset{\sim}{b}'\underset{\sim}{x} \quad \text{where } \underset{\sim}{b} = \frac{\underset{\sim}{a}}{\|\underset{\sim}{a}\|} \quad \text{and so } \|\underset{\sim}{b}\| = 1.$$

ORTHOGONAL PROJECTION OF $\underset{\sim}{x} \in \mathbb{R}^2$ ONTO $\underset{\sim}{a} \in \mathbb{R}^2$:

$$\hat{\underset{\sim}{x}} = d \cdot \frac{\underset{\sim}{a}}{\|\underset{\sim}{a}\|} = \frac{\underset{\sim}{a}\underset{\sim}{a}'}{\|\underset{\sim}{a}\|^2}\underset{\sim}{x} = \frac{\underset{\sim}{a}\underset{\sim}{a}'}{\underset{\sim}{a}'\underset{\sim}{a}} \cdot \underset{\sim}{x} = \underset{\sim}{b}\underset{\sim}{b}'\underset{\sim}{x}$$

ORTHOGONAL PROJECTION OPERATOR: $\dfrac{\underset{\sim}{a}\underset{\sim}{a}'}{\underset{\sim}{a}'\underset{\sim}{a}} = \underset{\sim}{b}\underset{\sim}{b}'$

LEMMA 7.1. FOR $\underset{\sim}{X}, \underset{\sim}{Y} \in \mathbb{R}^2$, $\underset{\sim}{X} \sim \underset{\sim}{Y}$ IFF $\underset{\sim}{a}'\underset{\sim}{X} \sim \underset{\sim}{a}'\underset{\sim}{Y}$ for all $\underset{\sim}{a} \in \mathbb{R}^2$,
OR EQUIVALENTLY IFF $\underset{\sim}{b}'\underset{\sim}{X} \sim \underset{\sim}{b}'\underset{\sim}{Y}$ for all $\underset{\sim}{b} \in \mathbb{R}^2$, $\|\underset{\sim}{b}\| = 1$.

THAT IS, A MULTIVARIATE DISTRIBUTION IS DETERMINED BY THE UNIVARIATE DISTRIBUTIONS OF ALL ITS UNIVARIATE PROJECTIONS.

PROOF: FOR $\underset{\sim}{X} \in \mathbb{R}^2$, THE CHARACTERISTIC FUNCTION $\phi_X(\underset{\sim}{s}) = E[e^{i\underset{\sim}{s}'\underset{\sim}{X}}]$ UNIQUELY DETERMINES F_X. THE CHARACTERITIC FUNCTION OF $W = \underset{\sim}{a}'\underset{\sim}{X}$ IS $\phi_W(t) = E(e^{itW}) = E[e^{it\underset{\sim}{a}'\underset{\sim}{X}}] = \phi_X(t\underset{\sim}{a})$.

HENCE $\phi_X(\underset{\sim}{s}) = \phi_{\underset{\sim}{s}'\underset{\sim}{X}}(1)$ WHICH UNIQUELY DETERMINES F_X. ∎

MOMENTS OF RANDOM VECTORS, $\underset{\sim}{X} \in \mathbb{R}^2$

MEAN VECTOR: $\underset{\sim}{\mu}_X = E[\underset{\sim}{X}] = \begin{bmatrix} E(X_1) \\ \vdots \\ E(X_2) \end{bmatrix} = \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_2 \end{pmatrix}$ IF IT EXISTS

NOTE: $\underset{\sim}{\mu}_X = \int \underset{\sim}{x}\, dF_X(\underset{\sim}{x}) \iff \mu_j = \int x_j\, dF_j(x_j)$. (WHY?)

2ND MOMENTS: COVARIANCE MATRIX (VARIANCE-COVARIANCE MATRIX)

$$\Sigma_X = \text{VAR}(\underset{\sim}{X}) \equiv E\left[(\underset{\sim}{X}-\underset{\sim}{\mu}_X)(\underset{\sim}{X}-\underset{\sim}{\mu}_X)'\right] = [\sigma_{ij}] \quad q \times q \text{ matrix}$$

diagonals: $\sigma_{ii} = E[(X_i-\mu_i)^2] = \text{VAR}(X_i) = \sigma_i^2$

off-diagonals: $\sigma_{ij} = E[(X_i-\mu_i)(X_j-\mu_j)] = \text{Cov}(X_i, X_j)$

PROVIDED $E[X_i^2]$ EXISTS FOR $i = 1, \ldots, q$.

NOTE: $\text{Cov}(X_i, X_j)$ MAY EXIST EVEN IF 2ND MOMENTS DO NOT EXIST.

IDENTITY: $\Sigma_X = E[\underset{\sim}{X}\underset{\sim}{X}'] - \underset{\sim}{\mu}\underset{\sim}{\mu}'$

EMPIRICAL DISTRIBUTION: SAMPLE $\{\underset{\sim}{x}_1, \ldots, \underset{\sim}{x}_n\}$ $\underset{\sim}{x}_i \in \mathbb{R}^p$ $i = 1, \ldots, n$

EMPIRICAL OR SAMPLE MEAN: $\bar{\underset{\sim}{x}} = \frac{1}{n}\sum_{i=1}^n \underset{\sim}{x}_i$

EMPIRICAL COVARIANCE MATRIX: $S_n = \frac{1}{n}\sum_{i=1}^n (\underset{\sim}{x}_i - \bar{\underset{\sim}{x}})(\underset{\sim}{x}_i - \bar{\underset{\sim}{x}})' = \frac{1}{n}\sum_{i=1}^n \underset{\sim}{x}_i \underset{\sim}{x}_i' - \bar{\underset{\sim}{x}}\bar{\underset{\sim}{x}}'$.

DATA MATRIX: $\underset{\sim}{X} = \begin{bmatrix} \underset{\sim}{x}_1' \\ \vdots \\ \underset{\sim}{x}_n' \end{bmatrix} = \begin{bmatrix} \underset{\sim}{x}_{(1)} & \cdots & \underset{\sim}{x}_{(q)} \end{bmatrix} = \{x_{ij}\}$ is $n \times q$

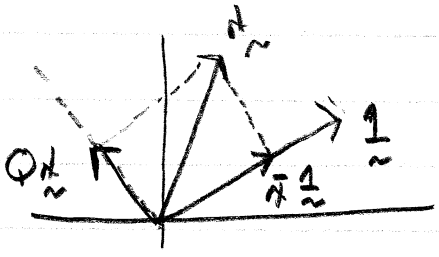
NOTE: $\underset{\sim}{x}_{(j)}$ represents a sample of size n from variable X_j.

So $\bar{\underset{\sim}{x}} = \frac{1}{n}\underset{\sim}{X}'\underset{\sim}{1}_n$ and $S_n = \frac{1}{n}\underset{\sim}{X}'\underset{\sim}{X} - \frac{1}{n^2}\underset{\sim}{X}'\underset{\sim}{1}_1\underset{\sim}{1}_n'\underset{\sim}{X} = \frac{1}{n}\underset{\sim}{X}'Q\underset{\sim}{X}$

where $Q = I_n - \frac{1}{n}\underset{\sim}{1}_n\underset{\sim}{1}_n'$

NOTE: $\frac{1}{n}\underset{\sim}{1}_n\underset{\sim}{1}_n'$ IS THE ORTHOGONAL PROJECTION ONTO $\underset{\sim}{1}_n$,

$Q = I_n - \frac{1}{n}\underset{\sim}{1}_n\underset{\sim}{1}_n'$ IS THE ORTHOGONAL PROJECTION ONTO THE $(n-1)$ dimensional subspace ORTHOGONAL TO $\underset{\sim}{1}_n$



$$Q\underline{X} = \begin{bmatrix} (\underset{\sim}{x}_1 - \bar{\underset{\sim}{x}})' \\ \vdots \\ (\underset{\sim}{x}_n - \bar{\underset{\sim}{x}})' \end{bmatrix}$$

Q: CENTERING MATRIX

NOTE: $Q = Q^2 = Q'$ (ORTHOGONAL PROJECTION MATRIX).

BASIC PROPERTIES OF $\underset{\sim}{\mu}_X$ AND Σ_X. $\underset{\sim}{X} \in \mathbb{R}^2$

LINEAR FUNCTION: $Y = \underset{\sim}{a}'\underset{\sim}{X} + b$ $\quad \underset{\sim}{a} \in \mathbb{R}^2, b \in \mathbb{R}$

i) $E(Y) = \underset{\sim}{a}'E(\underset{\sim}{X}) + b$ OR $\mu_Y = \underset{\sim}{a}'\underset{\sim}{\mu}_X + b$

ii) $\text{VAR}(Y) = \sigma_Y^2 = \underset{\sim}{a}'\Sigma_X\underset{\sim}{a}$ $\quad$ HW: VERIFY

LINEAR TRANSFORMATION: $\underset{\sim}{Y} = A\underset{\sim}{X} + \underset{\sim}{b}$ $\quad A_{p\times q}, \underset{\sim}{b} \in \mathbb{R}^p$

i) $E[\underset{\sim}{Y}] = AE[\underset{\sim}{Y}] + \underset{\sim}{b}$ OR $\underset{\sim}{\mu}_Y = A\underset{\sim}{\mu}_X + \underset{\sim}{b}$

ii) $\Sigma_Y = \text{VAR}(\underset{\sim}{Y}) = A\Sigma_Y A'$ $\quad$ HW: VERIFY

NOTE: WHEN A is $q\times q$ and nonsingular, this represents an affine transformation. The mean vector is said to be <u>affine invariant</u>.

LEMMA 7.2. IF Σ IS A COVARIANCE MATRIX, THEN Σ IS SYMMETRIC AND POSITIVE SEMI-DEFINITE ($\Sigma \geq 0$).

PROOF: SYMMETRY FOLLOWS SINCE $\text{Cov}(X_i, X_j) = \text{Cov}(X_j, X_i)$. TO SHOW $\Sigma \geq 0$, NOTE THAT FOR $\underset{\sim}{a} \in \mathbb{R}^q$, $\underset{\sim}{a}' \Sigma \underset{\sim}{a} = \text{VAR}(\underset{\sim}{a}' X) \geq 0$. ∎

DEFINITION 7.3, i) A MATRIX $A_{q \times q}$ IS POSITIVE DEFINITE IF AND ONLY IF $\underset{\sim}{a}' A \underset{\sim}{a} > 0$ for all $\underset{\sim}{a} \in \mathbb{R}^q$, $\underset{\sim}{a} \neq \underset{\sim}{0}$. NOTATION: $A > 0$.

ii) IT IS POSITIVE SEMI-DEFINITE (OR NON-NEGATIVE DEFINITE) IF and ONLY IF $\underset{\sim}{a}' A \underset{\sim}{a} \geq 0$ for all $\underset{\sim}{a} \in \mathbb{R}^q$. NOTATION: $A \geq 0$.

NOTES: i) THE MATRIX $A_{q \times q}$ NEED NOT BE SYMMETRIC.

ii) For a quadratic form over A,

$$g(\underset{\sim}{a}; A) = \underset{\sim}{a}' A \underset{\sim}{a} = \underset{\sim}{a}' A_s \underset{\sim}{a} = g(\underset{\sim}{a}; A_s) \quad \text{where } A_s = \tfrac{1}{2}(A + A')$$

is symmetric. HENCE, W.L.O.G. WE CAN ASSUME A IS SYMMETRIC.

iii) UNLESS OTHERWISE STATED, $A \geq 0$, $A > 0$ IMPLIES A IS ALSO SYMMETRIC.

LEMMA 7.4. $A > 0 \Rightarrow A^{-1} > 0$.

PROOF: A^{-1} EXISTS (VERIFY), and $\underset{\sim}{a}' A^{-1} \underset{\sim}{a} = \underset{\sim}{b}' A \underset{\sim}{b} > 0$, $\underset{\sim}{b} = A^{-1} \underset{\sim}{a}$.

LEMMA 7.5. i) $A_{q \times q} \geq 0$ IFF $A = BB'$ for some $B_{q \times q}$.

IN ADDITION, IF $\text{rank}(B) = q$ THEN $A > 0$.

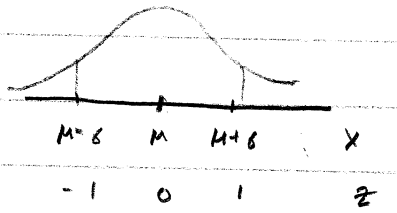
NOTE: B IS called a SQUARE ROOT OF A.

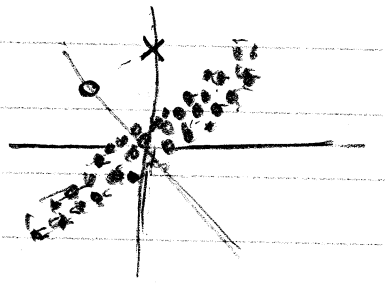
ii) IF B IS A SQUARE ROOT OF A, then C IS A SQUARE ROOT OF A IFF $C = BQ$ for some orthogonal $Q_{q \times q}$.

PROOF: LATER

INTERPRETATION OF $\underset{\sim}{\mu}$ and Σ.

1) $X \in \mathbb{R}$, $\mu, \sigma^2 \Rightarrow Z = \frac{X-\mu}{\sigma}$ STANDARD DEVIATION FROM MEAN





2) $\underset{\sim}{X} \in \mathbb{R}^2$, $\underset{\sim}{\mu}$, $\Sigma \Rightarrow$ FOR $\underset{\sim}{a}'\underset{\sim}{X}$, $Z_a = \frac{\underset{\sim}{a}'\underset{\sim}{X} - \underset{\sim}{a}'\underset{\sim}{\mu}}{\sqrt{\underset{\sim}{a}'\Sigma\,\underset{\sim}{a}}}$ FOR $\Sigma > 0$

PROBLEM: $\max_{\underset{\sim}{a} \in \mathbb{R}^2} |Z_a|$ ASSUME $\Sigma > 0$. EXPRESS $\Sigma = BB'$, B - full rank

SOLUTION: $\left[\underset{\sim}{a}'(\underset{\sim}{X}-\underset{\sim}{\mu})\right]^2 = \left\{\underset{\sim}{a}' B B^{-1}(\underset{\sim}{X}-\underset{\sim}{\mu})\right\}^2 = (\underset{\sim}{c}'\underset{\sim}{d})^2$ where $\underset{\sim}{c} = B'\underset{\sim}{a}$, $\underset{\sim}{d} = B^{-1}(\underset{\sim}{X}-\underset{\sim}{\mu})$

$\leq \underset{\sim}{c}'\underset{\sim}{c}\;\underset{\sim}{d}'\underset{\sim}{d}$ by CAUCHY-SCHWARTZ

$= \underset{\sim}{a}' BB' \underset{\sim}{a} \cdot (\underset{\sim}{X}-\underset{\sim}{\mu})'(BB')^{-1}(\underset{\sim}{X}-\underset{\sim}{\mu})$

$= \underset{\sim}{a}'\Sigma\,\underset{\sim}{a} \cdot (\underset{\sim}{X}-\underset{\sim}{\mu})'\Sigma^{-1}(\underset{\sim}{X}-\underset{\sim}{\mu})$

$\Rightarrow |Z_a| \leq D_{\underset{\sim}{\mu},\Sigma}(\underset{\sim}{X}) = \sqrt{(\underset{\sim}{X}-\underset{\sim}{\mu})'\Sigma^{-1}(\underset{\sim}{X}-\underset{\sim}{\mu})}$

$D_{\underset{\sim}{\mu},\Sigma}(\underset{\sim}{X})$: MAHALANOBIS DISTANCE WRT DISTRIBUTION F_X

NOTE: EQUALITY IS OBTAINED IFF $\underset{\sim}{c} \propto \underset{\sim}{d} \Leftrightarrow \underset{\sim}{a} \propto \Sigma^{-1}(\underset{\sim}{X}-\underset{\sim}{\mu})$

LEMMA 7.6. $\sup_{\underset{\sim}{a} \in \mathbb{R}^2} |Z_a| = D_{\underset{\sim}{\mu},\Sigma}(\underset{\sim}{X})$ OR EQUIVALENTLY

$\sup_{\underset{\sim}{b} \in \mathbb{R}^2,\ \|\underset{\sim}{b}\|=1} |Z_b| = D_{\underset{\sim}{\mu},\Sigma}(\underset{\sim}{X})$.